Reports – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Parent Module:** Reports · **Child Module:** 06.1 Reports **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

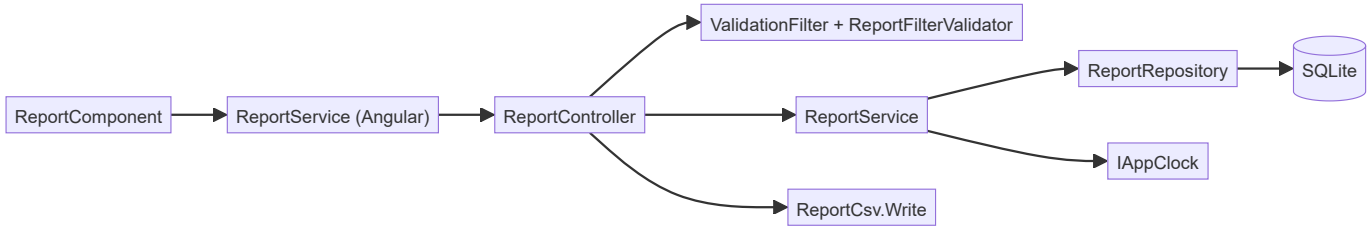
1. Technical Overview

The Reports child module implements BRD Screen 17. It has no tables of its own and reads the `Tasks`, `Users`, `UserProjectAllocations` and `Projects` tables. It exposes two GET endpoints under `api/v1/reports`:

- **Summary** – one aggregate query returns `TotalTasks`, `PendingTasks`, `CompletedTasks` and `OverdueTasks` for an optional filter (deadline range, assignee, project, status, priority).
- **Export** – the same filter gives up to 50,000 task rows. `ReportCsv` serialises them to RFC 4180 CSV with formula-injection protection and returns them as `taskflow-report.csv`.

Data is **organization-wide** for every caller who holds `reports.view`. No per-user scope is applied (design note D3 in `ReportService`). Soft-deleted tasks are excluded. Tasks of soft-deleted users are kept (`IgnoreQueryFilters`). The Angular `ReportComponent` is a standalone page. It reloads the summary on every filter change and downloads the CSV as a blob.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Files
Controller	`TaskFlow.Api/Controllers/Reports/`	`ReportController.cs`
Service	`TaskFlow.Api/Services/Reports/`	`IReportService.cs`, `ReportService.cs`, `ReportCsv.cs` (static CSV writer)
Repository	`TaskFlow.Api/Repositories/Reports/`	`IReportRepository.cs`, `ReportRepository.cs`, `ReportCriteria.cs` (record), `ReportTaskRow.cs` (record)
DTOs	`TaskFlow.Api/Dtos/Reports/`	`ReportDto.cs` (`ReportFilter`, `ReportSummaryDto`, `ReportRowDto`)
Validators	`TaskFlow.Api/Dtos/Reports/Validators/`	`ReportFilterValidator.cs`
Entities	`TaskFlow.Api/Models/Entities/`	`Task.cs`, `User.cs`, `UserProjectAllocation.cs`, `Project.cs` (read only)
Enums	`TaskFlow.Api/Models/Enums/`	`TaskStatus.cs`, `TaskPriority.cs`

Layer	Path	Files
Authorization	`TaskFlow.Api/Authorization/Policies/`, `TaskFlow.Api/Common/Security/`	`RequirePermissionAttribute.cs`, `PermissionPolicy.cs` (prefix `perm:`), `PermissionKeys.cs`, `PermissionCatalog.cs`
Cross-cutting	`TaskFlow.Api/Filters/`, `TaskFlow.Api/Middleware/`, `TaskFlow.Api/Authorization/Auditing/`	`ValidationFilter.cs`, `GlobalExceptionHandler.cs`, `DeniedActionAuditHandler.cs`
DI registration	`TaskFlow.Api/Config/`	`ServiceRegistration.cs` (`IReportService` → `ReportService`, scoped), `RepositoryRegistration.cs` (`IReportRepository` → `ReportRepository`, scoped)
Clock	`TaskFlow.Api/Services/Settings/ApplicationSettings/Clock/`	`IAppClock.cs`, `AppClock.cs` (`UtcNow`)
Help knowledge	`TaskFlow.Api/Services/Assistant/Knowledge/`	`reports-and-workload.md`

4. Frontend File Structure

Type	Path	Purpose
Page component	`src/app/modules/reports/pages/report.component.ts`, `report.component.html`	`ReportComponent` (selector `app-report`, standalone): filters, count cards, completion bar, Export CSV
Child components	–	None. Uses the shared `IconComponent` (`shared/components/icon/`)
API service	`src/app/modules/reports/services/report.service.ts`	`ReportService.getSummary(filter)`, `ReportService.exportCsv(filter)` (blob), `toParams()` drops empty values
Models	`src/app/modules/reports/models/report.model.ts`	`ReportFilter`, `ReportSummary` interfaces
Route / Guard	`src/app/modules/reports/reports.routes.ts`	`REPORTS_ROUTES`: path `reports`, `canActivate: [permissionGuard]`, `data.permissions: [Permissions.ReportsView]`. Loaded under the shell route in `src/app/routes/app.routes.ts` (`authGuard`, `mfaSetupGuard`)
Menu	`src/app/layouts/nav-items.ts`	"Reports" item (icon `chart`, route `/reports`) when `has(Permissions.ReportsView)`
Permission constants	`src/app/shared/auth/permissions.ts`	`ReportsView: 'reports.view'`, `ReportsExport: 'reports.export'`
Utilities used	`src/app/shared/utils/local-day.ts`, `src/app/shared/utils/save-blob.ts`, `src/app/shared/services/toast.service.ts`	Local-day to UTC bounds, blob download, error toast

Type	Path	Purpose
Cross-module services	`modules/management/project-management/services/project.service.ts` (`getAll`), `modules/monitoring/task-management/services/task.service.ts` (`getAssignees`)	Project and Employee dropdown options

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5. API & Logic

5.1 API Endpoints

#	Method	Endpoint	Description	Permission (Policy)	Request DTO	Response DTO
1	GET	`/api/v1/reports`	Totals and completion summary for the filtered scope	`reports.view` (class-level `[RequirePermission]`, policy `perm:reports.view`)	`ReportFilter` (query string)	`ReportSummaryDto` (200)
2	GET	`/api/v1/reports/export`	Filtered task rows as a CSV attachment	`reports.view` (class) **and** `reports.export` (`[Authorize(Policy = "perm:reports.export")]`)	`ReportFilter` (query string)	`text/csv` file `taskflow-report.csv` (200)

Query parameters of `ReportFilter` (all optional): `deadlineFrom` (DateTime), `deadlineTo` (DateTime), `assignedToUserId` (int), `projectId` (int), `status` (`TaskStatus` name: `ToDo`, `InProgress`, `Blocked`, `Completed`, `Cancelled`, `InQA`), `priority` (`TaskPriority` name: `Low`, `Medium`, `High`, `Critical`).

Dependency endpoints the screen calls (owned by other modules):

Method	Endpoint	Used for	Authorization
GET	`/api/v1/projects`	Project dropdown	`projects.view`
GET	`/api/v1/tasks/assignees`	Employee dropdown	`[Authorize]` only. The result is scoped to the caller's task access (`TaskService.GetAssigneesAsync`)

5.2 Request / Response Samples

```
// GET /api/v1/reports?deadlineFrom=2026-10-01T00:00:00.000Z&deadlineTo=2026-10-31T23:59:59.999Z&status=InProgress
// 200 OK
{ "totalTasks": 40, "pendingTasks": 28, "completedTasks": 12, "overdueTasks": 5 }
```

```
// 400 Bad Request (ValidationFilter)
{
  "type": "https://tools.ietf.org/html/rfc9110#section-15.5.1",
  "title": "One or more validation errors occurred.",
  "status": 400,
  "errors": { "DeadlineTo": ["The report 'to' deadline must be on or after the 'from' deadline."]
}
}
```

```
// GET /api/v1/reports/export?priority=High
// 200 OK  Content-Type: text/csv  Content-Disposition: attachment; filename=taskflow-report.csv
TaskCode,Title,Assignee,Project,Priority,Status,StartDate,Deadline,EstimatedHours,CompletedAt
TF-0012,Prepare Monthly Report,Asha Nair,"Apollo, Orion",High,Completed,2026-10-01,2026-10-05,6.5,2026-10-04
```

Lines end with CRLF. The values shown are illustrative. The task code format depends on the Task Management module.

5.3 Service Layer Logic

Class: `TaskFlow.Api.Services.Reports.ReportService : IReportService` (depends on `IReportRepository`, `IAppClock`). Constant `MaxExportRows = 50_000`.

Method: `GetSummaryAsync(ReportFilter filter, CancellationToken)`

1. Input is already validated by `ValidationFilter` with `ReportFilterValidator`.
2. Map `ReportFilter` → `ReportCriteria` through `Criteria(filter)`. Each deadline bound is normalised with `AsUtc`: a `Local` kind is converted with `ToUniversalTime()`, an `Unspecified` kind is treated as UTC (`SpecifyKind`), and a `Utc` kind is left unchanged.
3. Call `_reports.GetSummaryAsync(criteria, _clock.UtcNow, ct)`. "Overdue" is measured at the application clock's current UTC time.
4. Return the `ReportSummaryDto`. A filter that matches no tasks returns `(0, 0, 0, 0)`.

Method: `GetRowsAsync(ReportFilter filter, CancellationToken)`

1. Build `ReportCriteria` as above.
2. `_reports.GetRowsAsync(criteria, MaxExportRows, ct)` returns `ReportTaskRow` records ordered by `Deadline`, then `Id`, with at most 50,000 rows.
3. Collect the distinct `AssignedToUserId` values and call `_reports.GetProjectNamesByUserAsync(userIds, ct)`. Report filters are not applied here, so a deleted person's work keeps its projects.
4. Project each row to `ReportRowDto`. `Project` = `string.Join(", ", names)` of the assignee's allocated projects (sorted by name), or empty.
5. Return the list.

Static helper: `ReportCsv.Write(IEnumerable<ReportRowDto> rows)` (`ContentType = "text/csv"`, `FileName = "taskflow-report.csv"`)

1. Header row: `TaskCode,Title,Assignee,Project,Priority,Status,StartDate,Deadline,EstimatedHours,CompletedAt`, terminated by `\r\n`.
2. Each cell goes through `Escape`:
   - `null` becomes an empty string.
   - A value starting with `=`, `+`, `-` or `@` gets a leading `'` (formula-injection guard, D8).
   - A value containing `,`, `"`, CR or LF is wrapped in quotes, with inner quotes doubled.
3. Dates use `yyyy-MM-dd` (invariant culture). `EstimatedHours` uses invariant culture. `CompletedAt` is empty when null. `Priority` and `Status` are written as enum names.
4. The controller encodes the result as UTF-8 bytes (no BOM is added explicitly) and returns `File(bytes, "text/csv", "taskflow-report.csv")`.

No activity log, notification or mail is written by this module.

5.4 Repository Layer

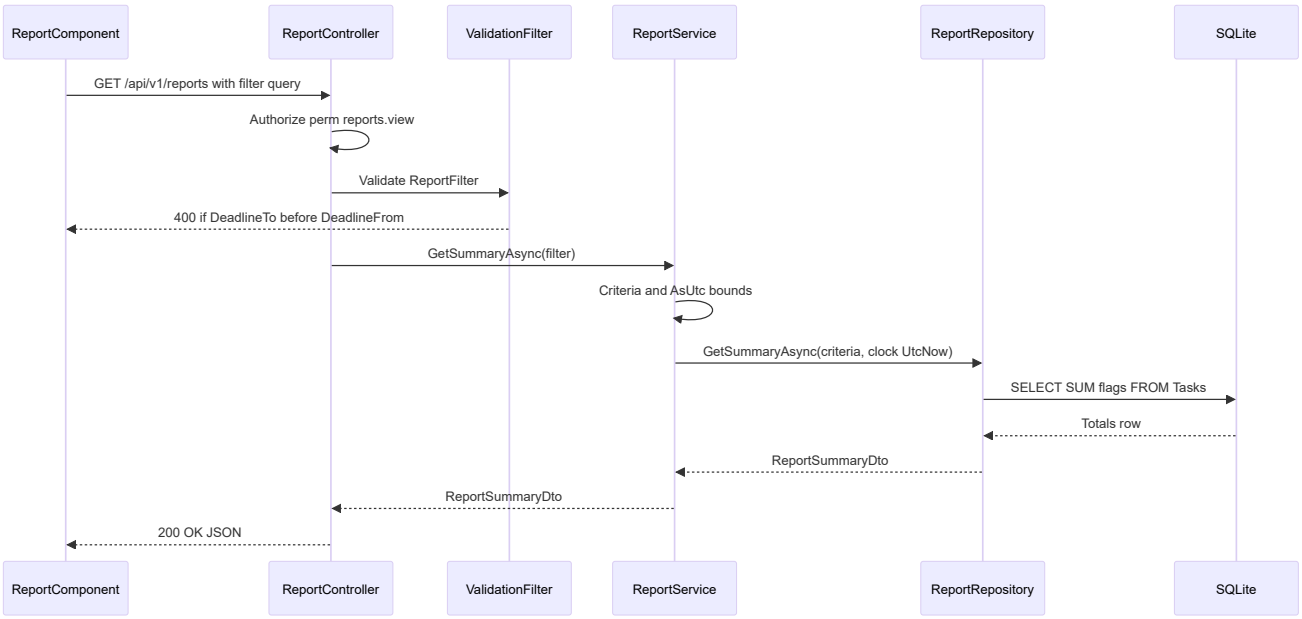
Class: `TaskFlow.Api.Repositories.Reports.ReportRepository : BaseRepository<Task>, IReportRepository`

Method	Description	Query Notes
`Covered(ReportCriteria)` (private)	Base query of covered tasks	`Set.AsNoTracking().IgnoreQueryFilters().Where(!task.IsDeleted)`. Optional filters: `Deadline >= DeadlineFromUtc`, `Deadline <= DeadlineToUtc` (both inclusive), `AssignedToUserId == x`, `AssignedTo.ProjectAllocations.Any(a => a.ProjectId == x)` (the **assignee's** allocation, not `Task.ProjectId`), `Status == x`, `Priority == x`
`GetSummaryAsync(criteria, now, ct)`	Totals in one aggregate query	Projects each task to 0/1 flags, `GroupBy(_ => 1)`, `Sum` per flag. Pending = status not `Completed` and not `Cancelled`. Completed = `Completed`. Overdue = pending and `Deadline < now`. `SingleOrDefaultAsync`. Returns zeros when there are no rows
`GetRowsAsync(criteria, take, ct)`	Export rows	`OrderBy(Deadline).ThenBy(Id).Take(take)`. Projects to `ReportTaskRow`. `Assignee` = `AssignedTo.FullName` or empty
`GetProjectNamesByUserAsync(userIds, ct)`	Project names per user	`Db.UserProjectAllocations.IgnoreQueryFilters().Where(userIds.Contains(UserId)).OrderBy(Project.Name)`. Groups in memory into `IReadOnlyDictionary<int, IReadOnlyList<string>>`

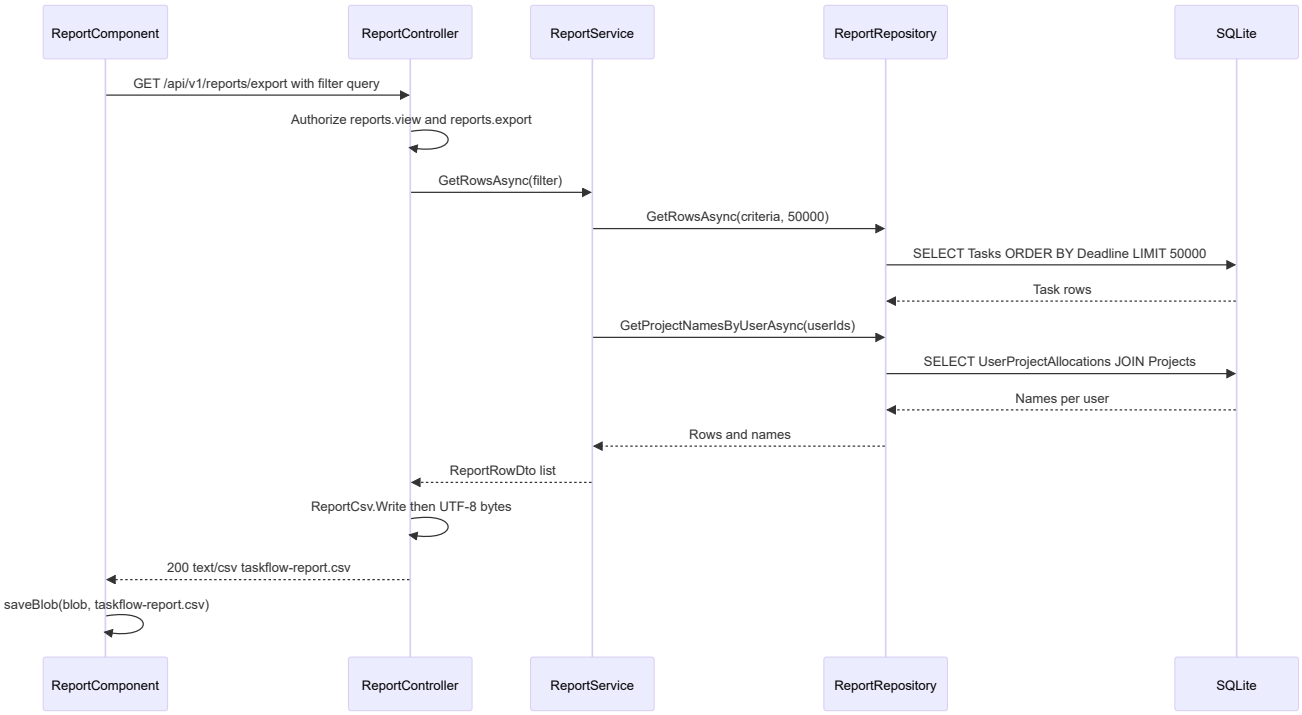
`IgnoreQueryFilters()` is used because `Users` has the soft-delete filter `!IsDeleted`, and reports must keep a deleted person's name and finished work.

5.5 Sequence Diagrams

Summary:



Export:



6. Database Design

This child owns no tables. It reads the following tables, named by their `AppDbContext` DbSets. Only the columns it uses are listed.

Table	Column	Type	Constraints	Description
`Tasks`	`Id`	int	PK	Tie-break sort for export
`Tasks`	`TaskCode`	string	Required, max 32, Unique index	CSV `TaskCode`
`Tasks`	`Title`	string	Required, max 200	CSV `Title`
`Tasks`	`AssignedToUserId`	int	FK → `Users.Id` (Restrict), indexed	Assignee filter, assignee name, project lookup
`Tasks`	`Priority`	`TaskPriority` (int)	–	Priority filter, CSV column
`Tasks`	`Status`	`TaskStatus` (int)	–	Status filter, Pending/Completed/Overdue rules
`Tasks`	`StartDate`	DateTime	–	CSV `StartDate`
`Tasks`	`Deadline`	DateTime	Indexed	Date-range filter, overdue rule, export sort
`Tasks`	`EstimatedHours`	decimal	–	CSV `EstimatedHours`
`Tasks`	`CompletedAt`	DateTime?	Nullable	CSV `CompletedAt`
`Tasks`	`IsDeleted`	bool	–	Deleted tasks are excluded

Table	Column	Type	Constraints	Description
`Tasks`	`ProjectId`	int?	FK → `Projects.Id` (Restrict)	**Not used** by reports
`Users`	`FullName`	string	Required, max 200	CSV `Assignee`
`Users`	`IsDeleted`	bool	Query filter `!IsDeleted` (bypassed by reports)	Soft delete
`UserProjectAllocations`	`UserId`, `ProjectId`	int, int	Composite PK. FKs to `Users` / `Projects` (Cascade). Query filter: user and project not deleted (bypassed)	Project filter, CSV `Project`
`Projects`	`Name`	string	Required, max 200, unique among non-deleted	CSV `Project` names

Relationships

- `Tasks.AssignedToUserId → Users.Id` (FK, Restrict)
- `Tasks.ProjectId → Projects.Id` (FK, Restrict). Not read by this child.
- `UserProjectAllocations.UserId → Users.Id` (FK, Cascade)
- `UserProjectAllocations.ProjectId → Projects.Id` (FK, Cascade)

Enum storage: `TaskStatus` = ToDo 0, InProgress 1, Blocked 2, Completed 3, Cancelled 4, InQA 5. `TaskPriority` = Low 0, Medium 1, High 2, Critical 3. They are stored as integers and sent over JSON as names (`JsonStringEnumConverter` in `CrossCuttingRegistration.cs`).

7. Security & Authorization

- **Authentication:** JWT bearer access token (`Jwt:AccessTokenMinutes`) plus a refresh-token cookie (`Jwt:RefreshTokenDays`).
- **Authorization:** Permission policies named `perm:` + key (`PermissionPolicy.NameFor`). `[RequirePermission(PermissionKeys.ReportsView)]` on the controller class covers both actions. `Export` adds `[Authorize(Policy = PermissionPolicy.Prefix + PermissionKeys.ReportsExport)]`, so both keys are required. Permission claims are refreshed from the database on every request. Refusals are logged by `DeniedActionAuditHandler` (user id, method, path, policy) and return the default 403.
- **Catalog (`PermissionCatalog.cs`):**
  - `reports.view`: "View reports", module `Report`, screen `Reports`, action `View`, `GrantedToManager: true`.
  - `reports.export`: "Export reports", action `Export`, `GrantedToManager: true`, `GrantFrom: reports.view`, `Requires: reports.view`.
  - Administrator is granted every key on every start (`AccessControlSeeder`). Employee has neither key by default.
- **Data scope: Organization** for every holder of `reports.view`. `ReportService` applies no Own/Team scope selector (D3, "matching the dashboards' scoping").
- **Frontend:**
  - `permissionGuard` on `/reports` (`reports.view`). A user without it is redirected to `dashboardRouteFor(role, permissions)`.
  - The nav item is shown via `has(Permissions.ReportsView)`.
  - The Export button is rendered only when `canExport()` (`authService.hasPermission(Permissions.ReportsExport)`).

Permission Key	Endpoint(s)	UI Element Controlled
`reports.view`	`GET /api/v1/reports`, `GET /api/v1/reports/export`	"Reports" menu item, `/reports` route, filters, count cards, completion bar
`reports.export`	`GET /api/v1/reports/export`	"Export CSV" button
`projects.view` (Management)	`GET /api/v1/projects`	Project filter options. Without it the call returns 403 and the dropdown stays empty, with no error shown

Role	`reports.view`	`reports.export`
Administrator	Granted	Granted
Manager	Granted	Granted
Employee	Denied	Denied
Custom Role	Configurable	Configurable

8. Validations & Error Handling

Rule	Layer	HTTP Code	Error Message
`DeadlineTo` earlier than `DeadlineFrom` (both supplied)	`ReportFilterValidator` via `ValidationFilter`	400	The report 'to' deadline must be on or after the 'from' deadline.
`Status` not a defined `TaskStatus`	`ReportFilterValidator` (`IsInEnum`)	400	FluentValidation default enum message
`Priority` not a defined `TaskPriority`	`ReportFilterValidator` (`IsInEnum`)	400	FluentValidation default enum message
Unparseable query value (e.g. bad date or unknown enum name)	MVC model binding (`[ApiController]`)	400	Default model-state validation problem
Not signed in / expired token	Authentication	401	–
Missing `reports.view` or `reports.export`	Policy	403	Forbidden (default handler, audited)
Unhandled fault	`GlobalExceptionHandler`	500	Something went wrong. Please try again.
Summary load fails (any error)	Frontend `ReportComponent.load`	–	Could not load the report. Check your connection and try again.
Export fails (any error)	Frontend `ReportComponent.exportCsv` (toast)	–	Could not export the report. Please try again.

Errors are returned as `application/problem+json` by `GlobalExceptionHandler`. Validation failures are returned as `ValidationProblemDetails` by `ValidationFilter`. No `AppException` (404/409) is thrown by this child. The frontend date inputs set `min`/`max` so the invalid range cannot normally be picked.

9. Integrations / Side Effects

- **Notifications:** None.
- **Email:** None.
- **Activity Log:** None. Only refused calls are logged, by `DeniedActionAuditHandler`.

- **Clock:** `IAppClock.UtcNow` (backed by `TimeProvider`) decides "overdue". The configured application timezone is **not** used for the deadline bounds. The browser converts local calendar days to UTC instants (`startOfLocalDay` / `endOfLocalDay`).
- **AI Assistant:** No report tool is exposed. The help article `Services/Assistant/Knowledge/reports-and-workload.md` (roles: Manager, Administrator, link `/reports`) describes the screen to users.

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`ConnectionStrings:Default`	SQLite database used by `AppDbContext`	`<secret>`
`Jwt:AccessTokenMinutes`	Access-token lifetime for calling the report APIs	30
`Jwt:RefreshTokenDays`	Refresh-token cookie lifetime	7
`Application:Timezone`	Application timezone read by `AppClock.GetTimeZoneAsync`. Not used by the report calculations	Asia/Kolkata
(code constant) `ReportService.MaxExportRows`	Export row cap	50000
(code constant) `ReportCsv.FileName` / `ContentType`	Export file name and media type	`taskflow-report.csv` / `text/csv`

The child has no configuration section of its own.

11. Testing

Test Type	File / Location	Covers
Unit	`tests/TaskFlow.UnitTests/Reports/ReportServiceTests.cs`	`GetSummaryAsync` totals (total, pending, completed, overdue) with no filter, combined filters and an empty scope; inclusive deadline bounds; a local-kind bound matches the same UTC instant; overdue uses the dashboard rule; priority and project (assignee's project) filters. `GetRowsAsync` uses the same filter as the totals and fills assignee, project, priority and status.
Unit	`tests/TaskFlow.UnitTests/Reports/ReportCsvTests.cs`	`ReportCsv.Write` puts the header row first, prefixes cells starting with `=`, `+`, `-` or `@` with `'`, quotes cells with commas, quotes or newlines, and writes dates (`yyyy-MM-dd`) and numbers in invariant format.
Integration	`tests/TaskFlow.IntegrationTests/Reports/ReportEndpointsTests.cs`	`GET /api/v1/reports`: Manager gets filtered totals, Administrator gets 200, Employee gets 403, combined filters narrow the totals, and `deadlineFrom` after `deadlineTo` returns 400. `GET /api/v1/reports/export`: Manager gets a `text/csv` attachment named `taskflow-report.csv` with the header row, rows match the summary filter, a title starting with `=` is exported with a leading `'` (formula-injection guard), and Employee gets 403.
Integration	`tests/TaskFlow.IntegrationTests/Common/PermissionAuthorizationTests.cs`	A custom role holding only `reports.view` gets 200 on `GET /api/v1/reports` and 403 on other gated areas.

Test Type	File / Location	Covers
Integration	`tests/TaskFlow.IntegrationTests/Common/ModuleBasedPermissionEndpointsTests.cs`	`GET /api/v1/reports/export` returns 403 without `reports.export` and is reached with `reports.export` plus `reports.view`.
Unit	`tests/TaskFlow.UnitTests/Settings/RoleConfiguration/PermissionCatalogTests.cs`	`reports.view` and `reports.export` sit in the "Report" module; old report access gains `reports.export` on upgrade; the Employee role does not hold `reports.view`.
Frontend unit	`src/app/modules/reports/pages/report.component.spec.ts`	`ReportComponent` shows the four totals and the completion summary, sends the selected filters (including `InQA`), downloads the CSV for the current filter, shows a toast when export fails, shows zeros for an empty scope, shows the error state with retry, and builds deadline bounds from local calendar days.
Frontend unit	`src/app/modules/reports/services/report.service.spec.ts`	`ReportService` sends `GET /api/v1/reports` with the supplied filters, leaves out empty filters, and downloads `GET /api/v1/reports/export` as a blob with the same filters.
Frontend unit	`src/app/shared/testing/permission-gating.spec.ts`	The Export CSV button is hidden for `reports.view` alone and shown with `reports.view` plus `reports.export`.
Frontend unit	`src/app/routes/app.routes.spec.ts`	The `reports` route uses `permissionGuard` with `reports.view` and loads `ReportComponent`.
Support	`tests/TaskFlow.UnitTests/TestSupport/WorkloadReportHarness.cs`	In-memory SQLite harness that builds `WorkloadService` and `ReportService` with a fake clock.

> `WorkloadReportHarness.cs` is shared. `ReportServiceTests.cs` uses it, and so do the workload tests `WorkloadServiceTests.cs` and `CustomRoleWorkloadTests.cs`.

12. Known Limitations / Gaps found

- The summary returns only four totals. There is no breakdown by status, priority or employee, and no chart library is used: the completion summary is a CSS progress bar.
- The export is capped at 50,000 rows. Larger results are cut off without warning, and the UI does not show that the cap was hit.
- The export is built fully in memory, as a string and then bytes, not streamed.
- The Employee and Project dropdown loads have no error handler. If those calls fail (e.g. 403 without `projects.view`), the dropdown stays empty with no message.
- The deadline-day conversion uses the browser's timezone, not `Application:Timezone`.
- Future: department filter, a per-user data scope, per-status breakdown and chart, and streamed or paged export.

Gaps found

- **BRD vs code – data scope:** BRD Screen 17 FR6 says "reports contain only records the user is permitted to access", but the code is **organization-wide** for any `reports.view` holder. The TDD (§22) records this as a known gap ("Reports (organization-wide, 50,000-row cap)").
- **BRD vs code – filter:** The BRD lists a **Department** filter. The code has a **Project** filter, and no department filter exists.
- **BRD vs code – chart:** The BRD lists a "Completion summary chart". The code renders four totals with a progress bar (recorded in the TDD as a simplified report summary).
- **BRD vs code – counts by status:** BRD FR2 "Display task counts by status" is only partly met (Total/Completed/Pending/Overdue).
- **Project semantics:** The project filter and CSV `Project` column use the **assignee's** project allocations, not `Tasks.ProjectId`. A task on project A, assigned to someone allocated to A and B, appears under both filters. A task whose assignee has no allocation is never matched by any project filter.
- **Deleted projects:** `GetProjectNamesByUserAsync` and the project filter use `IgnoreQueryFilters()`, so allocations to soft-deleted projects may still appear in the CSV `Project` column. ⚠️ To be confirmed: whether removing a project deletes its allocation rows or leaves them soft-filtered.
- **Hidden dependency:** The project filter needs `projects.view`, which is not declared in the catalog `Requires`. The Employee filter options come from `GET /api/v1/tasks/assignees`, which is scoped to the caller's task access. A custom role with `reports.view` but limited task access sees a partial employee list, yet reports over the whole organization.
- **Assistant knowledge vs code:** `reports-and-workload.md` says reports give "summaries of tasks by status, priority and employee". The code gives filtered totals only.

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